

Aniruddha Mukherjee

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RESEARCH INTERESTS

Computational genomics	Single-cell and spatial omics	Biological foundation models
Computational cancer biology	AI-driven protein design	Multi-omics integration
Software for biological research	Gene regulatory networks	AI for drug discovery

EDUCATION

Master of Technology, Biomedical Engineering Autumn 2025–Present

Biosciences and Bioengineering, Indian Institute of Technology Bombay, India

Advisors: Professor Shamik Sen and Professor Saket Choudhary

Bachelor of Technology, Computer Science and Engineering Autumn 2021–Spring 2025

School of Engineering, Tezpur University, India

Mentor: Professor Nityananda Sarma

PROJECTS

Seqout December 2025–Present

Search engine and metadata harmonisation for public genomic data across seven major repositories, with support for MCP tools for AI agents. Available via seqout.org.

Scinitilla September 2025–Present

Hardware-acceleration for single-cell RNA-seq count matrix normalisation using CUDA, MLX and Numba backends. Work currently private.

Tinypy 2024

Zero-dependency python package for linear algebra, compatible with NumPy, with methods for finding rank, eigenvalues and solving systems of equations. Available via [github](https://github.com).

Pathfinder 2024

GUI application written in Golang to visualise classic AI algorithms such as A* Search, Local Beam Search, Uniform Cost Search and more. Available via [github](https://github.com).

ByteCalcWeb 2024

Zero-dependency calculator with recursive descent parser implemented in TypeScript. Available via [github](https://github.com).

Lipika 2024

GPT-2 based small language model pre-trained in Assamese. Part of bachelor's thesis at Tezpur University.

Pygraphia 2023

Zero-dependency python package for graph data structures and algorithms, with methods for BFS, DFS and MST generation. Available via `pip` at pypi.org/project/pygraphia.

OCWD 2023

Command-line tool for downloading course material from MIT OpenCourseWare, at bulk. Written in Bash script. Over 1,000 downloads and 90 GitHub stars. Available via [github](https://github.com).

Fontonic

2023

Browser extension for customising typography of any website. Written in TypeScript. Over 25,000 daily users. Available via [github](#).

EXPERIENCE

Teaching Assistant

August 2026–Present

Autumn 2026: Programming for Biomedical Data Analysis (BB 709)

Research Intern

December 2024–January 2025

Department of Computer Science and Engineering, Indian Institute of Technology Patna

Mentor: Professor Rajiv Misra

Full-stack Engineering Intern

September 2020–November 2020

Foodable.in, New Delhi, India (Remote)

PUBLICATIONS

Aniruddha Mukherjee, Mukesh Reddy and Saket Choudhary “Seqout: metadata harmonisation for public genomic data discovery” (in submission; draft available upon request)

TALKS AND POSTER PRESENTATIONS

1. Aniruddha Mukherjee, Mukesh Reddy and Saket Choudhary. “Seqout: search engine for the world’s genomic data”. Talk presented at the 6th edition of IndiaFOSS, September 2026, Bengaluru, India. doi.org/10.5281/zenodo.23242105
 2. Aniruddha Mukherjee, Mukesh Reddy and Saket Choudhary. “Seqout: metadata harmonisation for public genomic data discovery”. Poster presented at:
 - The 6th Genomics India Conference, September 2026, Bengaluru, India doi.org/10.5281/zenodo.23239647
 - Mathematics of the real world: Research–Industry–Academia Workshop, August 2026, IIT Bombay, India. doi.org/10.5281/zenodo.23241736
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CERTIFICATIONS

- AWS Certified Cloud Practitioner. Issued by Amazon Web Services, August 2024.
- Fundamentals of Database Engineering. Issued by Udemy, June 2024.